Design of Protograph LDPC-Coded SCMA Systems With Power-Imbalanced Multi-Level Decoding

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***Abstract*—This paper studies the design of protograph low-density parity-check (PLDPC) coded sparse code multiple access (SCMA) systems with power-imbalanced multi-level decoding (PI-MLD) scheme. Specifically, by exploiting the protograph extrinsic information transfer (PEXIT) analysis, we first design two enhanced PLDPC (E-PLDPC) codes with the lowest decoding thresholds for the SCMA system over a Rayleigh fading channel. Furthermore, we conceive a new PI-MLD scheme, which is composed of level classification, power-imbalanced allocation (PIA), and power-oriented MLD, to further eliminate the SCMA inter-user interference. In particular, level classification and PIA can assign different powers to the users accessing the same resource, while the power-oriented MLD implements the sequential decoding according to a high-to-low power order among users. Analyses and simulation results show that the SCMA system with the proposed E-PLDPC codes and PI-MLD scheme is significantly superior to the prior art.**

***Index Terms*—Protograpgh low-density parity-check (PLDPC), power-imbalanced allocation (PIA), multi-level decoding (MLD), sparse code multiple access (SCMA), mutual information (MI).**

I. Introduction

Non-orthogonal multiple access (NOMA) technique allows multiple users (UEs) to access the same resources (REs), and thus is a promising solution for various 5G applications requiring high spectral efficiency and massive connectivity [1]. Particularly, the code-domain NOMA, such as *sparse code multiple access (SCMA)*, has attracted much attention in recent years due to its relatively high data-transmission reliability [2], [3], [4]. Specifically, the SCMA transmitter can map the binary bits to a sparse multi-dimensional complex modulation symbol (i.e., constellation point) [2]. The bit-to-symbol process helps obtain a large constellation shaping gain [5]. To enhance the flexibility of grouping UEs and supporting diverse data rates, a variable modulation SCMA (VM-SCMA) has been proposed in [3]. Furthermore, a novel nonlinear mapping has been designed to directly map the users' messages to a superimposed symbol, eliminating the need of a codebook for each UE [4]. On the other hand, the SCMA receiver can be implemented by the message passing algorithm (MPA), which can guarantee a near-optimal detection performance [2], [5].

For an SCMA system, error-correction codes (ECCs) are an effective solution to resist inter-user interference and enhance data-transmission reliability. In particular, as a type of powerful ECCs, low-density parity-check (LDPC) codes have been widely used in SCMA-related research works [6], [7]. Especially the irregular LDPC codes can significantly improve the SCMA system performance [7], [8]. However, the irregular LDPC codes suffer from relatively high encoding complexity. The structured LDPC codes, called *protograph LDPC (PLDPC) codes*, appear to a more desirable alternative for the SCMA system because they can keep the performance advantage while having a simpler implementation [9], [10], [11].

In parallel with the SCMA transmitter, there are a lot of research works on the SCMA receiver. For example, to reduce the computational complexity of the MPA, a modified version, namely log-domain MPA (log-MPA), has been considered [10]. Meanwhile, inspired by the expectation propagation (EP), an EP algorithm (EPA) has been exploited to further reduce the computational overhead [7]. However, in these works, the decoding procedures for different UEs are independent of one another, which cannot effectively eliminate the inter-user interference. To address this issue, the multi-level decoding (MLD), which has been widely used in the system with multiple data streams, may be a potential scheme [12]. This is because the MLD is a sequential decoding, in which the data stream decoded currently can help improve the decoding for subsequent data streams. In addition, we note that the power-imbalanced allocation (PIA) has been considered to make the data streams in an SCMA system have different convergence performance [13]. As such, incorporating the PIA into MLD may further enhance the performance of SCMA systems. In this paper, we focus on the design of PLDPC-coded SCMA systems with power-imbalanced MLD (PI-MLD). The main contributions of this paper are summarized as follows:

1) By exploiting the protograph extrinsic information transfer (PEXIT) algorithm [14], we design two different types of enhanced PLDPC (E-PLDPC) codes with the lowest decoding thresholds (i.e., having the strongest robustness to the inter-user interference) in the SCMA system over a Rayleigh fading channel.
2) Based on a given relationship between the UEs and REs (i.e., a given codebook), we propose a three-step design method consisting of level classification, PIA, and power-oriented MLD to construct a PI-MLD scheme, which further improves the overall SCMA system performance.
3) We conduct mutual-information analyses and bit-error-rate (BER) simulations to demonstrate that the SCMA system with the proposed E-PLDPC codes and PI-MLD scheme significantly outperforms the state-of-the-art counterparts.

II. PLDPC-Coded SCMA System Model

A. Research Motivation

SCMA systems with traditional LDPC codes suffer from relatively high encoding complexity, and existing decoding frameworks cannot effectively eliminate inter-user interference. To address the above issue, this paper first designs two low-complexity E-PLDPC codes with the lowest decoding thresholds by employing PEXIT algorithm, and then proposes a PI-MLD scheme with level classification, PIA, and power-oriented MLD to further enhance the overall system performance. In the remainder of this paper, we will first introduce the PLDPC-coded SCMA system model, and then put forward our proposed design.

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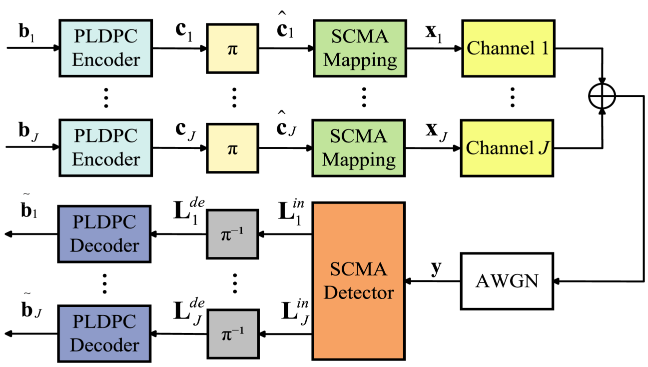

Fig. 1. Block diagram of an uplink PLDPC-coded SCMA system.

B. PLDPC-Coded SCMA System

Fig. 1 presents an uplink PLDPC-coded SCMA system diagram with J independent UEs and K orthogonal REs. In particular, the overloading factor is defined as $\lambda = J/K$ $(J > K)$, which indicates the system overloading capacity. Due to the SCMA sparsity, every UE accesses $d_v < K$ REs, while every RE is multiplexed by $d_f < J$ UEs. Additionally, the assignment relationship between J UEs and K REs is described by a factor graph matrix $\mathbf{F}_{K\times J}$ [2]. For example, the matrices $\mathbf{F}_{4\times 6}$ and $\mathbf{F}_{6\times 9}$ for both 4×6 and 6×9 SCMA systems are presented in (1). Specifically, the columns and the rows represent the UEs and the REs, respectively. Then, the elements '1' in the first column of $\mathbf{F}_{4\times 6}$ imply that UE 1 accesses REs 2 and 4, where $d_v = 2$ and $d_f = 3$.

$$\mathbf{F}_{4\times 6} = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \end{bmatrix},$$

$$\mathbf{F}_{6\times 9} = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}. \tag{1}$$

At the transmitter, the information-bit sequence $\mathbf{b}_j$ of UE $j \in \{1, 2, \ldots, J\}$ is encoded by a PLDPC encoder into a coded-bit sequence $\mathbf{c}_j$, which is then permuted by a random interleaver to be an interleaved version $\hat{\mathbf{c}}_j$. Subsequently, every $m = \log_2 M$ bits extracted from $\hat{\mathbf{c}}_j$ are mapped to a K-dimensional complex symbol $\mathbf{x}_j = [x_{1,j}, x_{2,j}, \ldots, x_{K,j}]^{\mathrm{T}}$ by the SCMA mapper, where $\mathbf{x}_j$ is one of the column vectors of the SCMA CB $\mathcal{X}^j$ with cardinality $|\mathcal{X}^j| = M$. At the SCMA receiver, the received signal vector $\mathbf{y} = [y_1, y_2, \ldots, y_K]^{\mathrm{T}}$ can be expressed as

$$\mathbf{y} = \sum_{j=1}^{J} \mathrm{diag}(\mathbf{h}_j)\mathbf{x}_j + \mathbf{n}, \tag{2}$$

where $\mathbf{h}_j = [h_{1,j}, h_{2,j}, \ldots, h_{K,j}]^{\mathrm{T}}$ denotes the channel gain vector and $\mathbf{n} \sim \mathcal{CN}(0, N_0)$ denotes a $K \times 1$ additive complex Gaussian noise vector, while N_0 is the noise power spectral density. For a given $\mathbf{y}$, the log-MPA algorithm [10] can be exploited to compute the bit-level extrinsic log-likelihood ratio (LLR) sequence $\mathbf{L}_j^{in}$, which is then processed by the deinterleaver to be a deinterleaved version $\mathbf{L}_j^{de}$. The $\mathbf{L}_j^{de}$ will be passed to the corresponding PLDPC decoder, and processed by the belief propagation (BP) algorithm [15]. Subsequently, the decoded information-bit sequence $\tilde{\mathbf{b}}_j$ can be yielded.

III. Proposed E-PLDPC Codes and PI-MLD Scheme for SCMA System

A. Propoesd E-PLDPC Codes

A protograph with a code rate $R = (n_v - n_c)/n_v$ has n_c check nodes (CNs), n_v variable nodes (VNs), and n_e edges connecting the two different types of nodes [15]. Such a protograph can be represented by a base matrix $\mathbf{B} = (b_{i,j})$ of size $n_c \times n_v$, where $b_{i,j}$ is the number of edges connecting CN c_i and VN v_j. Based on the modified progressive-edge-growth (PEG) [16] algorithm, one can extend the base matrix to a parity check matrix of size $wn_c \times wn_v$, which is used to generate the corresponding rate-R PLDPC code. In particular, w denotes the lifting factor.

A given PLDPC code usually has different performance behavior in different communication scenarios [15]. Then, the existing excellent PLDPC codes designed for the single-user system may not work well in the multi-user system, such as the SCMA system. To improve the performance of the SCMA system over a Rayleigh fading channel, we propose two new PLDPC codes, including an unpunctured E-PLDPC code and a punctured E-PLDPC code.

1) Unpunctured E-PLDPC Code: As is well known, the degree-2 VNs can help ensure a low decoding threshold for PLDPC codes. However, excessive degree-2 VNs will make the PLDPC codes suffer from a severe error floor in the high SNR region [17]. Besides, based on the PEXIT-aided computer-search method, choosing a large initial protograph can increase the probability of searching a good PLDPC code, but it also increases the computational complexity. According to the above discussion, we consider a 4×8 initial base matrix with 32 elements and employ the following constraints.

a) *Error-Floor Elimination:* To avoid the error floor, the number of VNs with degrees not greater than 2 should be limited by $n_c - 1$ [18] ($n_c = 4$ in the initial base matrix).

b) *Low Decoding Threshold:* To ensure a low decoding threshold, we assume one degree-1 VN, one highest-degree VN, and two degree-2 VNs, which correspond to the VNs numbered 1 to 4, respectively. In particular, for the 4-th VN, we initialize $b_{4,4} = 1$. Thus, only one of $b_{1,4}$, $b_{2,4}$, and $b_{3,4}$ will be 1, while the other two will be 0.

c) *Low Computational Complexity:* To reduce the code-search space, all elements within the base matrix are assumed to be no larger than 3 ($b_{i,j} \in \{0, 1, 2, 3\}$, $i = 1, 2, \ldots, 4$ and $j = 1, 2, \ldots, 8$). Also, we set three degree-3 VNs ($\sum_{i=1}^{4} b_{i,5} = \sum_{i=1}^{4} b_{i,6} = \sum_{i=1}^{4} b_{i,7} = 3$) and one second highest-degree VN satisfying $3 < \sum_{i=1}^{4} b_{i,8} < \sum_{i=1}^{4} b_{i,2}$.

The initial base matrix $\mathrm{B}_{\mathrm{U}}^{0}$ for a rate-$1/2$ PLDPC code is expressed in (3). Particularly, the second column and the eighth column are the highest-degree VN and the second highest-degree VN, respectively. After a simple computer search, the resultant base matrix $\mathrm{B}_{\mathrm{U}}^{1}$ with the lowest decoding threshold can be obtained and called *Code-U*.

$$\mathrm{B}_{\mathrm{U}}^{0} = \begin{bmatrix} 1 & b_{1,2} & 0 & b_{1,4} & b_{1,5} & b_{1,6} & b_{1,7} & b_{1,8} \\ 0 & b_{2,2} & 1 & b_{2,4} & b_{2,5} & b_{2,6} & b_{2,7} & b_{2,8} \\ 0 & b_{3,2} & 1 & b_{3,4} & b_{3,5} & b_{3,6} & b_{3,7} & b_{3,8} \\ 0 & b_{4,2} & 0 & 1 & b_{4,5} & b_{4,6} & b_{4,7} & b_{4,8} \end{bmatrix},$$

$$\mathrm{B}_{\mathrm{U}}^{1} = \begin{bmatrix} 1 & 3 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 1 & 1 & 1 & 0 & 0 & 3 & 0 \\ 0 & 2 & 1 & 0 & 1 & 2 & 0 & 1 \\ 0 & 2 & 0 & 1 & 2 & 1 & 0 & 3 \end{bmatrix}. \tag{3}$$

2) Punctured E-PLDPC Code: Compared to unpunctured PLDPC codes, the punctured PLDPC codes have a high potential to exhibit better performance. Furthermore, we find that introducing a unique node connection method (also called pre-coding structure) [15] can significantly improve the coding performance. Specifically, the pre-coding structure consists of one degree-1 VN, one highest-degree punctured VN, and one associated CN only connecting the former two VNs [15], [18]. Therefore, we design a rate-$1/2$ punctured PLDPC code, in which the first two columns and the first row constitute the pre-coding structure. The initial 4×7 base matrix $\mathrm{B}_{\mathrm{P}}^{0}$ with one degree-1 VN, one highest-degree punctured VN, two degree-2 VNs, and three degree-3 VNs, which respectively correspond to the VNs 1 to 7, is presupposed as

$$\mathrm{B}_{\mathrm{P}}^{0} = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & b_{2,2} & 1 & b_{2,4} & b_{2,5} & b_{2,6} & b_{2,7} \\ 0 & b_{3,2} & 1 & b_{3,4} & b_{3,5} & b_{3,6} & b_{3,7} \\ 0 & b_{4,2} & 0 & 1 & b_{4,5} & b_{4,6} & b_{4,7} \end{bmatrix},$$

$$\mathrm{B}_{\mathrm{P}}^{1} = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 & 2 & 2 & 0 \end{bmatrix}. \tag{4}$$

After the computer search, the resultant base matrix $\mathrm{B}_{\mathrm{P}}^{1}$ of the punctured PLDPC code, which has the lowest decoding threshold (referred to as Code-P), can be obtained.

Remark 1:

- For the initialization procedure of $\mathrm{B}_{\mathrm{U}}^{0}$ and $\mathrm{B}_{\mathrm{P}}^{0}$, distributing several '1' elements in different rows is to ensure that the resultant $\mathrm{B}_{\mathrm{U}}^{1}$ and $\mathrm{B}_{\mathrm{P}}^{1}$ achieve low decoding thresholds while avoiding the cycle-4 structure.
- During the PEXIT analysis, both SCMA property and Rayleigh fading channel characteristic are considered in the Monte-Carlo simulation, which provides the LLRs of channel outputs to derive the corresponding decoding threshold of the PLDPC code.
- This work focuses on the design of rate-$1/2$ E-PLDPC codes. For other code rates, separate base matrices can be designed by using PEXIT analysis. The proposed Code-U and Code-P have the lowest decoding thresholds and can achieve optimal performance with a code rate of $1/2$ across different information lengths based on a well-designed lifting algorithm [16].
- The proposed E-PLDPC codes are designed based on a 4×6 CB-1 [19] and the log-MPA [10] (without using MLD). In Section III-B, we will explore the integration of MLD and further conduct an investigation on power allocation to improve the overall system performance.

B. Proposed PI-MLD Scheme

As a sequential decoding scheme, the MLD can use the soft information of the data stream decoded currently to improve the decoding performance of the subsequent data streams [12]. In order to maximize the MLD performance gain in SCMA systems, it is essential to ensure that different data streams (i.e., different UEs) have different reliabilities. For this reason, PIA has a high potential to enhance the MLD performance. In this subsection, we propose the PI-MLD scheme, which is a three-step design consisting of level classification, PIA, and power-oriented MLD.

1) Level Classification: To ensure the normalized energy for different REs to be equal in the SCMA mapping [13], we classify all J UEs into L different levels by imposing the following two constraints: (i) The numbers of UEs classified in the L levels are the same L^U, i.e., $J = L \times L^U$; (ii) In every level, the L^U UEs must occupy all K REs, while the numbers of UEs assigned in different REs are the same, i.e., $L^U \times d_v \bmod K = 0$. In particular, the constraint (i) helps determine the parameters L and L^U, while the constraint (ii) is to determine the classification relationship between the J UEs and the L levels.

Based on the above two constraints, for an SCMA system with $\mathbf{F}_{4\times6}$, the $J = 6$ UEs are classified into $L = 3$ levels. Particularly, the levels 1, 2, and 3 are composed of UEs 1, 2, UEs 3, 4, and UEs 5, 6, respectively. Similarly, the $J = 9$ UEs in the SCMA system with $\mathbf{F}_{6\times9}$ can also be classified into $L = 3$ levels. The levels 1, 2, and 3 are composed of UEs $1 \sim 3$, UEs $4 \sim 6$, and UEs $7 \sim 9$, respectively. Additionally, level classification for the SCMA system with different factor graphs is shown in Table I.

TABLE I
LEVEL CLASSIFICATION FOR DIFFERENT SCMA FACTOR GRAPHS

K REs $\times$ J UEs	$\lambda = J/K$	Number of levels L	UE number for each level L^U
5×10	2	2	5
4×6, 6×9, 8×12	1.5	3	2, 3, 4
6×8, 9×12	1.33	4	2, 3

Algorithm 1: Proposed PIA Optimization.

1 **Input**: Given parameters L, $p_n = 1, n \in \{1, 2, \cdots, L\}$, $\delta = 0.025$, calculate AMI I_{AMI} using modified PEXIT, $I_{old} = 0$.
2 **for** $i = 1$ ***to*** $\lfloor L/2 \rfloor$ **do**
3   **for** $iteration = 1$ ***to*** $\lfloor 1/\delta \rfloor - i$ **do**
4   **if** $i = 1$ ***or*** $p_i < (p_{i-1} - \delta)$ **then**
5       $p_{L+1-i} = p_{L+1-i} - \delta$; $p_i = p_i + \delta$;
6   Calculate I_{AMI} using the PEXIT.
7   **if** $p_i \leq 1 + \delta \times (\lceil L/2 \rceil - i)$ **then**
8   **continue**;
9   **if** $I_{AMI} > I_{old}$ **then**
10         $I_{old} = I_{AMI}$;
11   **else**
12         $p_{L+1-i} = p_{L+1-i} + \delta$; $p_i = p_i - \delta$;
13 **if** $L \mod 2 \neq 0$ **then**
14   Repeat steps $3 \sim 13$ for $p_{\lceil L/2 \rceil}$ and p_1, and keep $p_{\lceil L/2 \rceil} > p_{\lceil L/2 \rceil + 1}$;
15 **Output**: The locally optimal power vector $\mathbf{p}_o = [p_1, p_2, \cdots, p_L]$.

2) PIA: After level classification, we aim to optimize the PIA scheme for the UEs classified into L different levels. To be specific, we propose Algorithm 1 exploiting a modified PEXIT analysis to search the locally optimal PIA scheme. The modified PEXIT analysis can simulate the MLD process, and track the convergence performance for every UE from the average mutual information (AMI) perspective. Before implementing Algorithm 1, we impose five initial constraints: (i) The total transmit power for the entire SCMA system is kept constant; (ii) The transmit powers for different UEs within the same level are the same, but for the UEs within different levels are different; (iii) Assuming the J UEs in an SCMA system are classified into L different levels, the corresponding power vector $\mathbf{p} = [p_1, p_2, \ldots, p_L]$ must satisfy the conditions $p_1 > p_2 > \cdots > p_L > 0$ and $p_n \neq 1$, where $n = 1, 2, \ldots, L$; (iv) When L is an even number, the number of low-power levels is $L^{low} = L/2$ (i.e., $p_{\frac{L}{2}+1}, p_{\frac{L}{2}+2}, \ldots, p_L$); Otherwise (i.e., L is an odd number), $L^{low} = \lceil L/2 \rceil$; (v) The minimum step size for the power adjustment is set to $\delta = 0.025$.

During the PIA optimization process given in Algorithm 1, the energy of all UEs should be first normalized. Then, in the i-th ($i = 1, 2, \ldots, \lfloor L/2 \rfloor$) iteration, one can continually increase the power of the UEs within the i-th level p_i and accordingly reduce the power of the UEs within the $L + 1 - i$ level p_{L+1-i}, while keeping the power of the UEs within the remaining levels (i.e., p_n for $n \neq i$ and $n \neq L + 1 - i$) constant, until the AMI converges to a locally optimal value. The above operation will be repeated for different level pairs. Moreover, the power

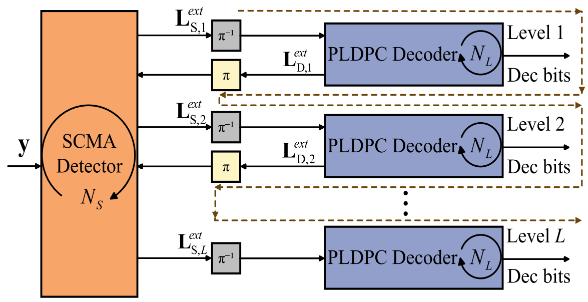

Fig. 2. The block diagram of the proposed power-oriented MLD scheme.

adjustment for different UEs within a given level is the same. When L is an even number, the PIA optimization process is terminated after completing the last level pair $p_{\frac{L}{2}}$ and $p_{\frac{L}{2}+1}$. If L is an odd number, the power of the UEs within the ($\lceil L/2 \rceil$)-th level $p_{\lceil L/2 \rceil}$ is reduced, while accordingly increasing p_1 so as to ensure the optimal system performance. Consequently, the locally optimal power vector $\mathbf{p}_o$ can be obtained.

Remark 2: The PIA is a continuously varying optimization problem because the power can be any positive value. It is then almost impossible to search for the globally optimal power vector. However, as the step size δ decreases, the resultant power vector can approach the globally optimal solution.

3) Power-Oriented MLD: Fig. 2 shows the block diagram of the proposed power-oriented MLD scheme. The π^{-1} and π denote the de-interleaver and interleaver, respectively. The $\mathbf{L}^{ext}_{\mathrm{S},i}$ (resp. $\mathbf{L}^{ext}_{\mathrm{D},i}$) denotes the *extrinsic* bit LLRs passed from the SCMA detector to the PLDPC decoder (resp. passed from the PLDPC decoder to the SCMA detector) in the i-th level, where $i = 1, 2, \ldots, L$. The maximum number of detection (resp. decoding) is set to N_S (resp. N_L). The dotted line indicates the decoding order for all L different levels. Specifically, after the first-round SCMA detection, the PLDPC decoders corresponding to the UEs with the highest power p_1 will perform the BP decoding first, and then feed back relatively accurate *extrinsic* LLRs to the SCMA detector. These feedback LLRs of the UEs with p_1 and the initial LLRs of the remaining UEs serve for the second-round SCMA detection. Subsequently, the PLDPC decoders corresponding to the UEs with the second highest power p_2 perform the BP decoding and feed back their *extrinsic* LLRs used for next SCMA detection. This sequential update composed of SCMA detection and BP decoding is implemented for different UEs based on a high-to-low power order (instead of a low-to-high power order) to guarantee a significant performance gain, and is terminated when the BP decoding on the UEs with the lowest power p_L is completed.

Remark 3: The design of E-PLDPC codes and PIA are independent to each other. The former aims to simultaneously and equally enhance the performance for all UEs in the SCMA system, while the latter focuses on working in conjunction with MLD. The UEs with p_1 ensure a good performance due to a large power, while the UEs with p_i ($i = 2, 3, \ldots, L$) attain a significant performance gain based on a reliable inter-user interference cancellation in SCMA detection.

IV. Numerical Results

In this section, we present the numerical analyses on the SCMA systems over a Rayleigh fading channel. Particularly, the proposed E-PLDPC codes (Code-U and Code-P) are adopted, while the $(3, 6)$ [17], repeat-jagged-accumulate (RJA) [20], new radio (NR) [21], Code-NOMA [22], Code-M2 [23], and accumulate-repeat-by-4-jagged-accumulate (AR4JA) [17] codes are used as benchmarks. All these PLDPC codes have the same code rate $1/2$ and information-bit length 1200. The 4×6 CB-1 [19], CB-2 [24], and 5×10 CB-3 [25], CB-4 [24] are considered, and the modulation order is set to $M = 4$. Furthermore, the SCMA system, MLD-SCMA system (without PIA), and PI-MLD-SCMA system are taken into account. Similar to the existing studies [26], [27], the maximum iteration numbers of the log-MPA and the BP are set to $N_S = 5$ and $N_L = 50$, respectively.

TABLE II
Decoding Thresholds (SNR dB) of the Proposed Code-U and Code-P, and the Existing $(3, 6)$, RJA, NR, and AR4JA Codes in the SCMA System

Unpunctured Codes			Punctured Codes		
Code-U	RJA [20]	$(3, 6)$ [17]	Code-P	NR [21]	AR4JA [17]
4.833	5.138	5.253	4.770	4.876	4.880

The 4×6 CB-1 [19] is considered.

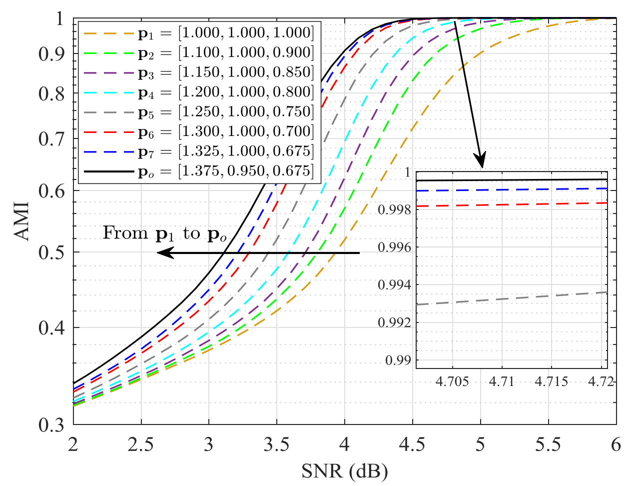

Fig. 3. The AMI convergence curves of the proposed PI-MLD-SCMA system with CB-1 and different PIA schemes.

A. Convergence-Performance Analysis

By exploiting the PEXIT analysis, we calculate the decoding thresholds of the proposed Code-U and Code-P, unpunctured $(3, 6)$ and RJA, punctured NR and AR4JA. Referring to Table II, where the SCMA system with CB-1 is considered, the proposed Code-U has a lower decoding threshold 4.833 dB than both $(3, 6)$ and RJA, while the proposed Code-P achieves a lower decoding threshold 4.770 dB than both NR and AR4JA. Also, we analyze the AMI convergence behavior of the proposed PI-MLD-SCMA system with Code-P and CB-1 during the PIA optimization process. In Fig. 3, one can note that the AMI will gradually converge to a local optimum (i.e., the power vector varies from $\mathbf{p}_1 = [1, 1, 1]$ to $\mathbf{p}_o = [1.375, 0.950, 0.675]$ with the aid of the PIA optimization given in Algorithm 1). This implies that the PI-MLD-SCMA system with $\mathbf{p}_o$ possesses the best error performance. Note that the proposed PIA optimization is general and applicable to the SCMA systems with various factor graphs and CBs. As a further insight, Table III shows the decoding thresholds of UEs within different levels in the MLD-SCMA and proposed PI-MLD-SCMA systems, where the proposed Code-P, CB-1, and CB-3 are used. One can easily find that for the proposed PI-MLD-SCMA system with CB-1, the UEs in level 1 exhibit the lowest decoding threshold 2.953 dB, while the UEs

TABLE III
DECODING THRESHOLDS (SNR DB) OF UES WITHIN DIFFERENT LEVELS IN THE MLD-SCMA AND PROPOSED PI-MLD-SCMA SYSTEM

SCMA \ CB	4 × 6 CB-1			5 × 10 CB-3	
	Level 1	Level 2	Level 3	Level 1	Level 2
MLD	4.770	4.368	4.112	7.057	6.391
PI-MLD	2.953	3.401	3.640	4.981	5.772

The proposed Code-P, 4 × 6 CB-1 [19], and 5 × 10 CB-3 [25] are used.

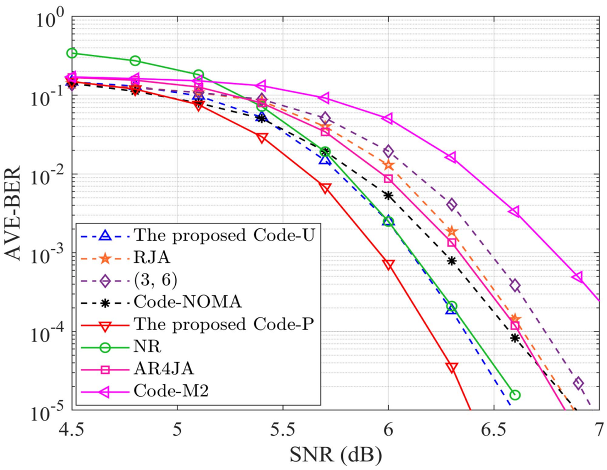

Fig. 4. AVE-BER curves of the unpunctured codes (Code-U, $(3, 6)$, RJA, Code-NOMA) and punctured codes (Code-P, NR, AR4JA, Code-M2) in the SCMA system with CB-1.

in level 2 have a lower decoding threshold 3.401 dB than those in level 3. Also, a similar decoding-threshold phenomenon can be found from CB-3. Besides, it is obvious that the decoding thresholds of all UEs in the proposed PI-MLD-SCMA system are better than those in the MLD-SCMA system.

B. Error-Rate Performance

Fig. 4 depicts the average BER (AVE-BER) curves of the unpunctured codes (Code-U, $(3, 6)$, RJA, Code-NOMA) and punctured codes (Code-P, NR, AR4JA, Code-M2) in the SCMA system with 4×6 CB-1. As observed from the dashed lines, at an AVE-BER of 1×10^{-5}, the proposed Code-U obtains performance gains of about 0.3, 0.3, and 0.45 dB over the $(3, 6)$, RJA, and Code-NOMA, respectively. The solid lines also illustrate that the proposed Code-P significantly outperforms the NR, AR4JA, and Code-M2.

Another important measure for PLDPC codes is decoding complexity expressed as $\mathcal{O}(N_L(n\bar{V} + (n-k)\bar{C}))$ [28], where $\bar{V}$ (resp. $\bar{C}$) denotes the average degree of VNs (resp. CNs), k and $n-k$ denote the information-bit length and check-bit length, respectively. Based on the base matrices of Code-P and NR, the decoding complexity of Cod-P $\mathcal{O}(N_L(6.0n))$ is significantly lower than that of NR code $\mathcal{O}(N_L(7.0n))$.

Fig. 5 shows the performance of proposed PI-MLD-SCMA system with locally optimal power vector $\mathbf{p}_o$. In particular, the CB-1, CB-2, CB-3, CB-4, and Code-P are used. Referring to Fig. 5(a), at an AVE-BER of 1×10^{-5}, the PI-MLD-SCMA system with CB-1 (CB-2) and corresponding $\mathbf{p}_o = [1.375, 0.950, 0.675]$ ($\mathbf{p}_o = [1.350, 0.950, 0.700]$) can achieve a performance gain of about 1.6 dB (1.5 dB). Similarly, the PI-MLD-SCMA system with CB-3, CB-4, and corresponding $\mathbf{p}_o = [1.375, 0.625]$ achieves a performance gain

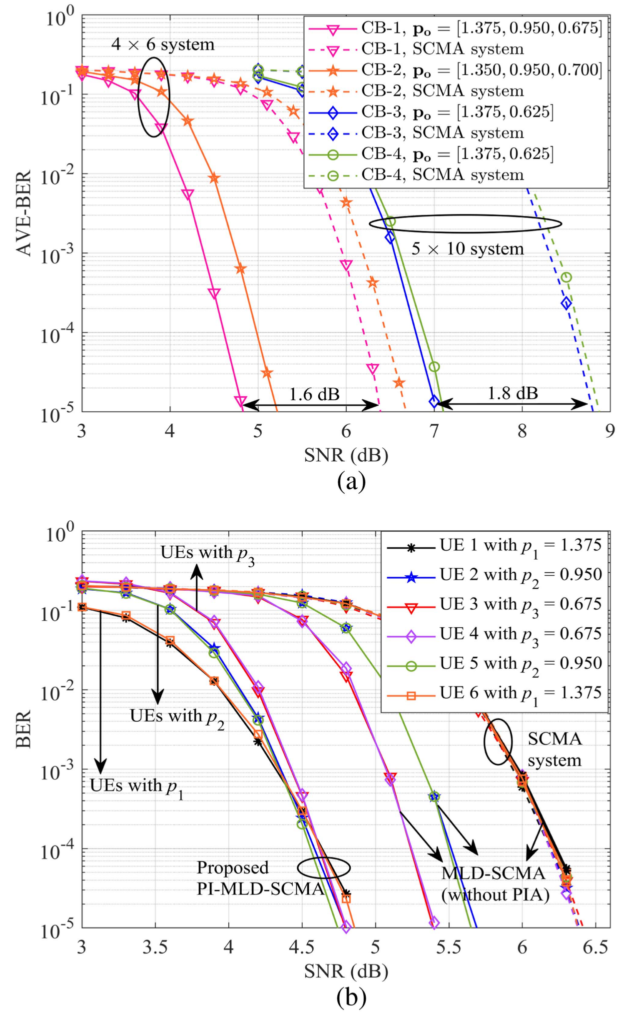

Fig. 5. (a) AVE-BER curves of the SCMA and proposed PI-MLD-SCMA systems with the CB-1, CB-2, CB-3, CB-4, and Code-P; (b) BER curves of all UEs in the SCMA, MLD-SCMA, and proposed PI-MLD-SCMA systems with the CB-1 and Code-P.

of about 1.8 dB. Fig. 5(b) shows the BER curves for all UEs (instead of AVE-BER) in the SCMA, MLD-SCMA, and proposed PI-MLD-SCMA systems, while the Code-P and CB-1 are used. Obviously, one can observe that at a BER of 1×10^{-5}, all UEs in the proposed PI-MLD-SCMA system can significantly outperform those in the SCMA and MLD-SCMA systems. Note also that for the MLD-SCMA system, the users belonging to the 1-st level cannot benefit from the power allocation and the MLD procedure, and thus have the same performance as in the SCMA system.

Remark 4: In Fig. 5(b), we assume that UEs 1, 6, UEs 2, 5, and UEs 3, 4 correspond to levels 1, 2, and 3, respectively. Actually, the mapping relationship between the UEs and the levels is not fixed. Any UE can be chosen as level 1.

To further validate the superiority of proposed PI-MLD design, Fig. 6 shows the performance of the SCMA system with power-imbalanced CB (PI-CB) and power $\mathbf{p}_1$ [13], and the proposed PI-MLD-SCMA system with CB-1, CB-2 and the corresponding locally optimal power vector $\mathbf{p}_o$. For both SCMA systems, the Code-P is used as the channel coding scheme. As seen in the figure, the proposed PI-MLD-SCMA system with CB-1 (resp. CB-2) and $\mathbf{p}_o = [1.375, 0.950, 0.675]$ (resp.

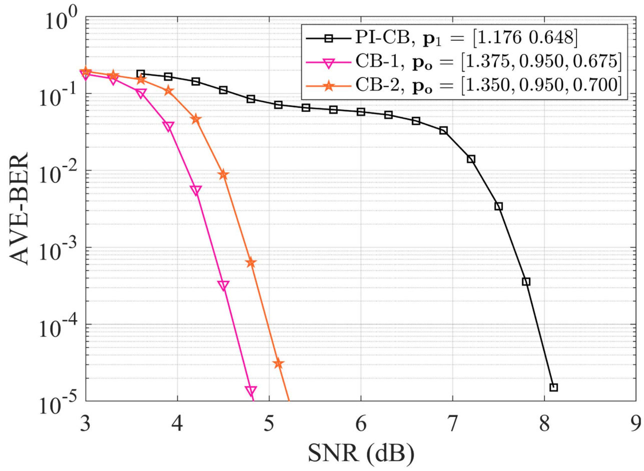

Fig. 6. AVE-BER curves of the SCMA system with PI-CB and power $\mathbf{p}_1$ [13], and the proposed PI-MLD-SCMA system with CB-1, CB-2 and the corresponding locally optimal power vector $\mathbf{p}_o$. Also, the Code-P is used as the channel coding scheme.

$\mathbf{p}_o = [1.350, 0.950, 0.700]$) outperforms the SCMA system with PI-CB and power $\mathbf{p}_1 = [1.176, 0.648]$.

V. Conclusion

This paper has investigated the PI-MLD-SCMA system with E-PLDPC codes over a Rayleigh fading channel. Specifically, we have first exploited the PEXIT algorithm to construct two different types of E-PLDPC codes (unpunctured Code-U and punctured Code-P) having the lowest decoding thresholds in the SCMA system. Subsequently, we have conceived a three-step design method composed of level classification, PIA, and power-oriented MLD to develop a new PI-MLD scheme for the SCMA system. Numerical analyses have shown that our proposed designs significantly outperform state-of-the-art counterparts, and thus stand out as a potential transceiver framework for SCMA applications with high-reliability requirements. As a future line, we will extend this work to more general and practical SCMA with large-scale fading. Besides, we will explore the integration of PIA with dynamic power optimization to further enhance the system performance.

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